CS383 · SOFTWARE ENGINEERING

Lecture 6 Project Management

Midterm summary

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01 Software project management

It is the coordination and management of the tasks and the engineers who work on the project.

Success criteria

- The software is delivered at the agreed time.
- The cost stays within the budget.
- The software meets the specification.
- The team is motivated, comfortable, and works well.

The manager's tasks

Task	What the manager does
Planning	Estimates the cost and schedules the project, assigns people to tasks, and tracks the progress.
Reporting	Writes reports on the progress.
Risk management	Identifies the risks, assesses them, and acts if a problem occurs.
People management	Chooses the team members and helps them work effectively.
Proposal writing	Writes a proposal to win a contract: the objectives, and the cost and schedule estimates.

02 Project planning

The plan includes: milestones (**Milestones**), breaking the problem down into tasks, scheduling the tasks, and estimating the time and resources. The plan is documented in the **SPM**.

Contents of the plan

Section	What it contains
Introduction	The objectives and the constraints (time and budget).
Project organization	How the team is organized and the members' roles.
Risk analysis	The risks and the ways to reduce them.
Resource requirements	The hardware and software needed for the development.
Work breakdown	Breaking the work down into activities and milestones.
Project schedule	The dependencies between the activities, the time of each milestone, and the allocation of people.

Good to know

We reduce the dependencies between tasks to avoid delays. Planning tools include Jira and Microsoft Project.

03 Estimation techniques

There are two ways to estimate the effort:

1 EXPERIENCE-BASED By experience

The manager uses experience from previous projects. Its weakness: the new project may not be similar to the old one, and without strong experience the estimate becomes inaccurate.

2 ALGORITHMIC COST MODELING By formula

```
Effort = A × Size^B × M
```

Symbol	Meaning
`A`	A constant that depends on the organization and the type of software.
`Size`	The size of the code or an estimate of the functionality.
`B`	The size and complexity of the system. Its value is between `1` and `1.5`.
`M`	A multiplier that combines the characteristics of the process, the product, and the team.

Estimate accuracy

The initial estimate is not accurate. If the estimate is `x`, the actual value ranges from `0.25x` to `4x`. The estimate improves as the project progresses.

04 Software pricing

It is presenting the cost of the project to the customer. We estimate the effort for each activity, then calculate the cost, then the price.

Cost items

```
Effort costs            (salaries of engineers and managers)
```

```
Hardware and software costs, including maintenance
```

```
Travel and training costs
```

Other factors that affect the price

Factor	How it affects the price
Market opportunity	A low price so that we can enter a new market and become known. The profit comes later.
Contractual terms	The customer lets the company keep ownership of the code and use it in other projects.
Requirements volatility	If the requirements are likely to change, we offer a lower price at the start and charge for the changes later.
Financial health	A company in financial difficulty lowers the price, because cash flow matters more than profit.

05 People management

Four factors the manager cares about

Factor	Meaning
Consistency	Everyone feels that their contribution is valued fairly.
Respect	The manager respects everyone, however different their skills are.
Inclusion	Everyone feels that the others listen to them.
Honesty	The manager is always honest.

Teamwork

A large project is not given to one large team, because time is lost in communication and the tasks become hard to track. Small teams (fewer than 10) are better:

- Fewer communication problems.
- The team is cohesive (**Cohesive**) and the members know each other.
- It is easier to develop quality standards.
- It is easier to encourage refactoring and continuous improvement.

06 Meetings

A successful meeting has: a clear goal, an **Agenda** (the list of topics), a start time and an end

time, and **Minutes** (a record of the decisions).

Type	Goal	Notes
Status meeting	Everyone knows where the project stands.	Short, from 10 to 15 minutes. At least once a week.
Brainstorming session	We come up with many solutions to a problem.	We do not criticize any idea during the session. The evaluation comes at the end.
Decision-making meeting	We choose between alternatives.	Rare. The participants must know the alternatives before the meeting.

Brainstorming and Decision-making meetings are called **Action meetings**.

Project control

We track the work against the plan and correct it if there is a deviation. Adding people to a late project makes it later.

07 Risk management

It is anticipating the risks that could affect the project's schedule or its quality.

Types of risk

Type	Affects	Example
Project risk	The schedule or the resources.	Essential hardware does not arrive on time.
Product risk	The quality or performance of the product.	The size of the system was underestimated.
Business risk	The organization itself.	A competing product reaches the market before we finish.

Risk management steps

#	Step	What it produces
1	Risk identification	List of potential risks
2	Risk analysis	Prioritized risk list
3	Risk planning	Risk avoidance and contingency plans
4	Risk monitoring	Risk assessment

Risk analysis scales

Scale	Values
Probability	very low · low · moderate · high · very high
Impact	catastrophic · serious · tolerable · insignificant

Examples from the slides

Risk	Plan for dealing with it
Financial problems reduce the project budget.	We prepare a report for management that shows the importance of the project and why cutting the budget is not a good idea.
A key staff member falls ill at a critical time.	We organize the team so that the work overlaps and people understand each other's work.
It is hard to recruit people with the required skills.	We warn the customer about a possible delay, and we look into buying ready-made components.

08 Common mistakes

Mistake	Correct
Saying that adding people speeds up a late project	It makes it later.
Confusing Project risk and Product risk	**Project** is the schedule and the resources. **Product** is the quality and the performance.
Putting B outside its range	`B` is between 1 and 1.5.
Saying that the initial estimate is accurate	The actual value is from 0.25x to 4x.
Saying that the Status meeting is long	It is short, 10 to 15 minutes.

09 Test yourself

1. Write the formula for Algorithmic cost modeling.
2. "A competing product reaches the market before we finish" is which type of risk?
3. List the four factors in People management.
4. What are the four steps of risk management, in order?
5. Why are small teams better? Give two reasons.

Answers

#	Answer
1	Effort = A × Size^B × M
2	**Business risk**.
3	**Consistency**, **Respect**, **Inclusion**, **Honesty**.
4	Identification, then Analysis, then Planning, then Monitoring.
5	Fewer communication problems, and the team is cohesive.